

# GDP360 Database Design & Schema



The GDP360 Database Schema Document is an extension of the GDP360 Hackathon initiative. It provides the complete structural blueprint of the relational database that powers the GDP360 analytical ecosystem — designed to capture, organize, and analyze 50 years of macroeconomic indicators across multiple countries.

# Document Purpose & Objectives

## This document serves two purposes:

- To explain the logical data model supporting GDP360 analytics.
- To guide participants in translating conceptual data relationships into SQL-based schema creation and KNIME integration.

## Core Objectives

- Establish a normalized and scalable relational schema for GDP and macroeconomic indicators.
- Define key entities, relationships, and constraints for consistent data modeling.
- Enable seamless data integration and analysis across Country, Year, and Economic Metrics dimensions.
- Provide the foundation for KNIME workflows, SQL-based exploration, and BI visualization.



# Business Context

## Analytical Scope

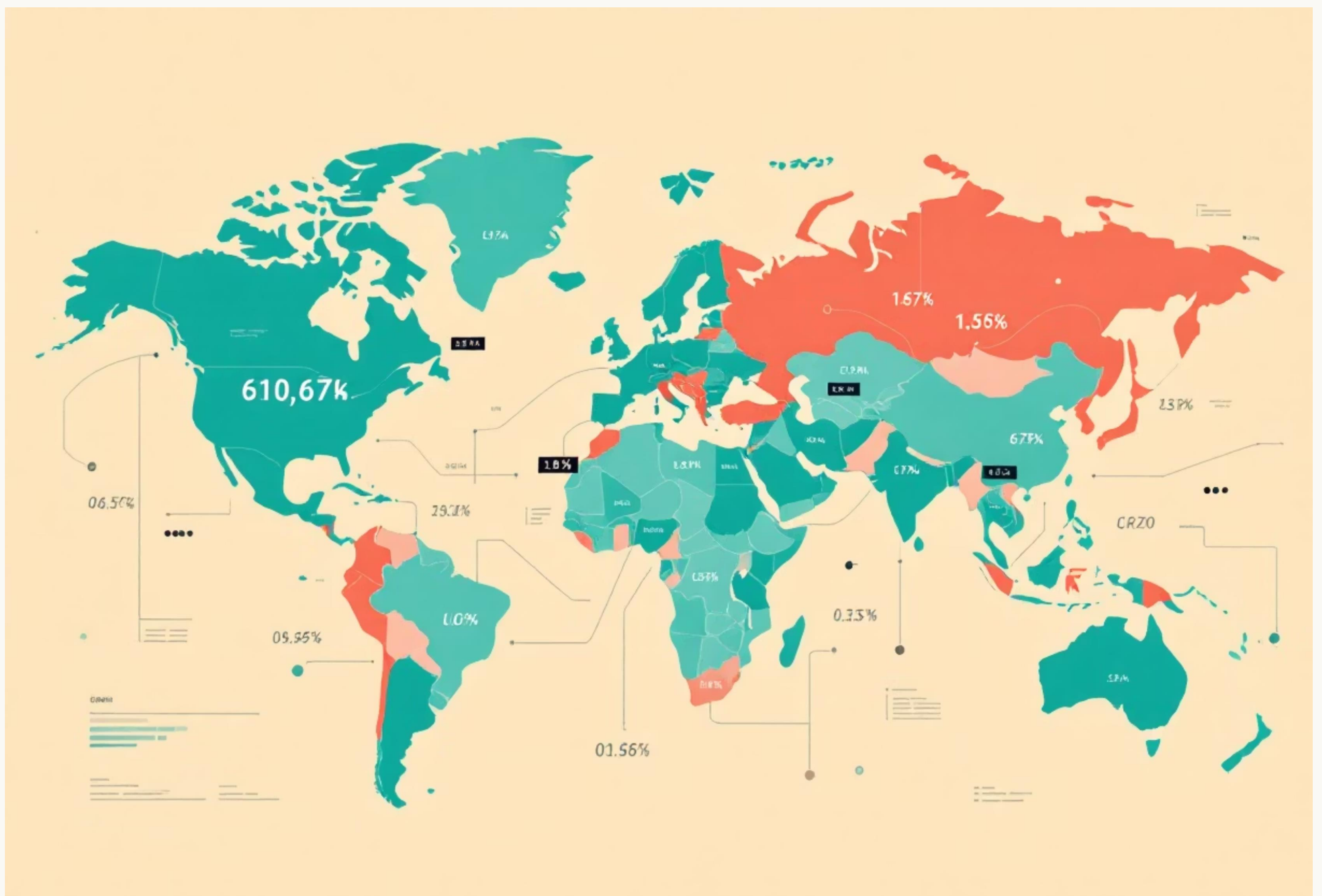
The GDP360 database is designed for analytical exploration of global growth patterns between 1975–2025.

## Data Consolidation

It consolidates essential indicators such as GDP, Population, Electricity Access, Unemployment, and Rural Population.

## Strategic Insights

Allows users to analyze economic evolution, compare country performance, and derive policy insights.



# Scope of the System



## Country Master

Stores metadata of all countries (ID, name, ISO code).



## Year Master

Defines the timeline of available data (1975–2025).



## GDP Metrics

Captures GDP values per country and year.



## Electricity Access

Tracks percentage of population with electricity access.



## Population Statistics

Stores population figures for each country-year pair.



## Unemployment Rate

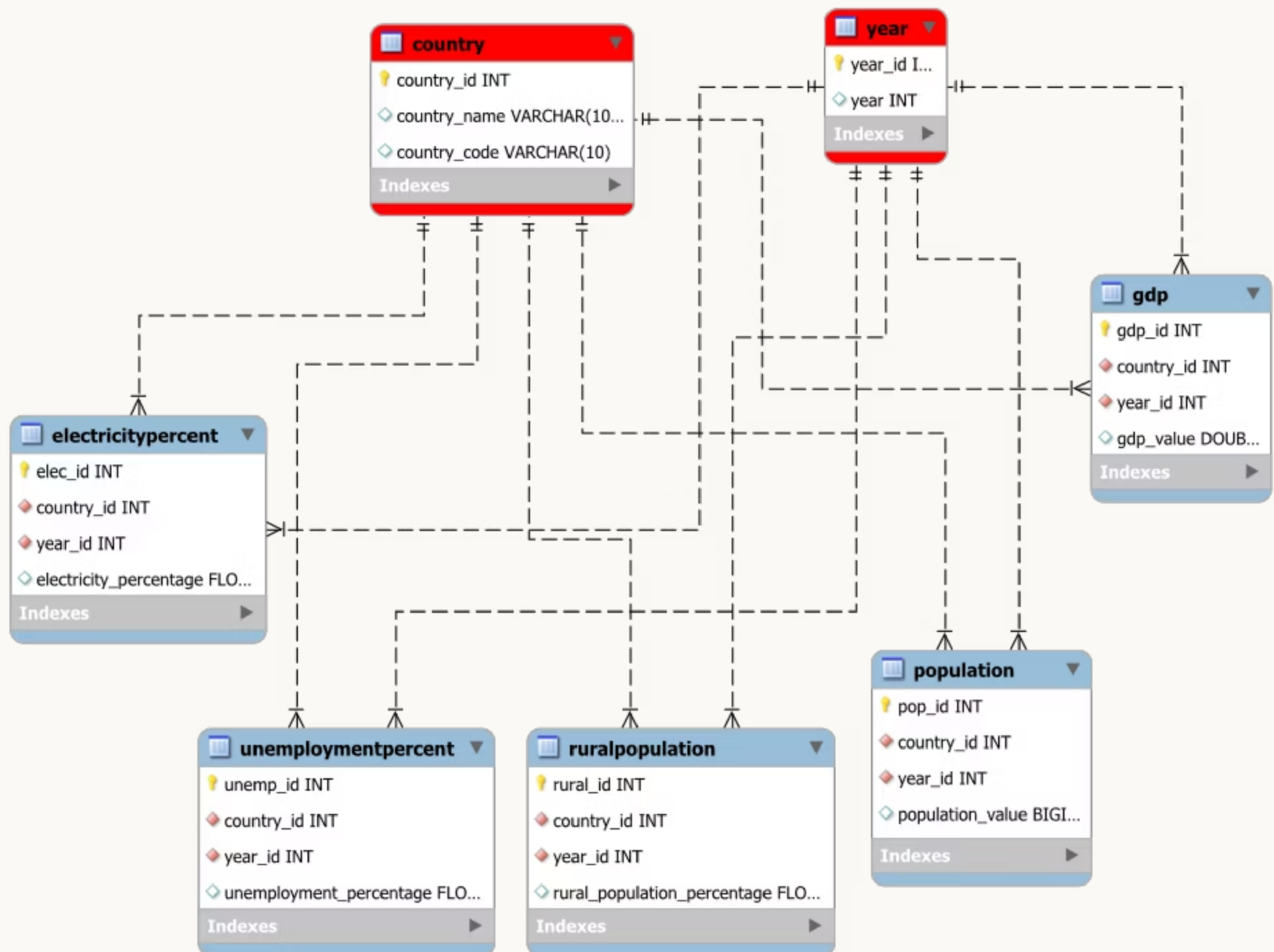
Records unemployment percentages.



## Rural Population

Captures rural population share as a percentage of total.

# Database Schema Overview



The schema diagram above illustrates the complete relational structure of the GDP360 database, showing how master tables (Country and Year) connect to all indicator fact tables through foreign key relationships.



# Common Tables Overview

| Table Name          | Columns                                                                            | Primary Key | Foreign Key                             |
|---------------------|------------------------------------------------------------------------------------|-------------|-----------------------------------------|
| Country             | country_id (PK),<br>country_name,<br>country_code                                  | country_id  | -                                       |
| Year                | year_id (PK), year                                                                 | year_id     | -                                       |
| GDP                 | gdp_id (PK),<br>country_id (FK),<br>year_id (FK),<br>gdp_value                     | gdp_id      | country_id → Country,<br>year_id → Year |
| ElectricityPercent  | elec_id (PK),<br>country_id (FK),<br>year_id (FK),<br>electricity_percentage       | elec_id     | country_id → Country,<br>year_id → Year |
| Population          | pop_id (PK),<br>country_id (FK),<br>year_id (FK),<br>population_value              | pop_id      | country_id → Country,<br>year_id → Year |
| UnemploymentPercent | unemp_id (PK),<br>country_id (FK),<br>year_id (FK),<br>unemployment_percentage     | unemp_id    | country_id → Country,<br>year_id → Year |
| RuralPopulation     | rural_id (PK),<br>country_id (FK),<br>year_id (FK),<br>rural_population_percentage | rural_id    | country_id → Country,<br>year_id → Year |



# Individual Table Definitions

## Master Tables

Table: Country

| Column Name  | Data Type    | Constraint            |
|--------------|--------------|-----------------------|
| country_id   | INT          | PRIMARY KEY, NOT NULL |
| country_name | VARCHAR(100) | NOT NULL              |
| country_code | VARCHAR(10)  | UNIQUE, NOT NULL      |

Table: Year

| Column Name | Data Type | Constraint            |
|-------------|-----------|-----------------------|
| year_id     | INT       | PRIMARY KEY, NOT NULL |
| year        | INT       | UNIQUE, NOT NULL      |

## Economic Indicator Tables



**Table: GDP**

**Columns:** gdp\_id (PK), country\_id (FK), year\_id (FK), gdp\_value

**Data Type:** gdp\_value as DECIMAL(18,2)



**Table: ElectricityPercent**

**Columns:** elec\_id (PK), country\_id (FK), year\_id (FK), electricity\_percentage

**Data Type:** electricity\_percentage as DECIMAL(5,2)



**Table: Population**

**Columns:** pop\_id (PK), country\_id (FK), year\_id (FK), population\_value

**Data Type:** population\_value as BIGINT



**Table: UnemploymentPercent**

**Columns:** unemp\_id (PK), country\_id (FK), year\_id (FK), unemployment\_percentage

**Data Type:** unemployment\_percentage as DECIMAL(5,2)



**Table: RuralPopulation**

**Columns:** rural\_id (PK), country\_id (FK), year\_id (FK), rural\_population\_percentage

**Data Type:** rural\_population\_percentage as DECIMAL(5,2)

# Relationships & Constraints

# Table Relationships

| Relationship Type | Parent Table | Child Table          | Description                                           |
|-------------------|--------------|----------------------|-------------------------------------------------------|
| 1:N               | Country      | GDP                  | One country can have many GDP records (one per year). |
| 1:N               | Country      | Population           | One country can have multiple population entries.     |
| 1:N               | Country      | ElectricityPercent   | Electricity access varies yearly.                     |
| 1:N               | Country      | UnemploymentPercent  | Unemployment changes yearly.                          |
| 1:N               | Country      | RuralPopulation      | Rural data tracked over time.                         |
| 1:N               | Year         | All indicator tables | Each year maps to several country records.            |

## Constraints & Integrity Rules

## Foreign Key Requirements

All fact tables must include both `country_id` and `year_id` as foreign keys.

## Uniqueness Constraint

Each (country\_id, year\_id) pair must be unique in its respective fact table.

## Referential Integrity

Referential integrity ensures that no data exists without a corresponding country and year entry.



# Use Cases & Analytical Scenarios



## GDP Growth Trend

Compute YoY growth for each country using GDP table.



## Electricity vs. GDP

Correlate electricity access with GDP growth.



## Population Impact

Compare population growth with GDP per capita.



## Employment Analysis

Relate unemployment rate trends to GDP performance.



## Rural Shift

Observe rural-to-urban transition patterns over decades.



# Integration & Future Enhancements

## Integration with KNIME & BI Tools

The GDP360 database acts as the foundational data source for the KNIME Analytical Workflow. Once structured, the schema supports:

- KNIME data import and ETL nodes (File Reader, Joiner, GroupBy).
- Analytical transformations (GDP per capita, CAGR, correlation analysis).
- Visualization and dashboard generation in KNIME, Power BI, or Tableau.

## Future Enhancements



### Additional Indicators

Include additional macro indicators (Inflation, Debt, CO<sub>2</sub> Emissions).



### Machine Learning

Implement time-series forecasting using KNIME Machine Learning nodes.



### Regional Aggregation

Extend schema to include Regional Aggregation Tables.



### BI Dashboards

Build Power BI or Tableau dashboards using this schema as the backend.